Week2 Exercise: LLM prompt engineering problems

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Introduction

Large Language Models (LLMs) have become valuable tools for generating content, analyzing text, and solving complex problems. However, the quality and accuracy of their responses depend heavily on how instructions are provided. Prompt engineering is the practice of designing and refining prompts to guide LLMs toward producing more relevant, accurate, and structured outputs. Effective prompt engineering techniques such as role prompting, output constraints, task decomposition, and context specification can significantly improve model performance across a wide range of applications. In this exercise, I applied different prompt engineering strategies to text generation, sentiment analysis, and mathematical reasoning tasks to evaluate how well carefully designed prompts influence the quality, precision, and consistency of AI-generated results. This paper discusses the prompts used, the outcomes achieved, and the lessons learned from the exercise.

LLM Prompt Engineering Problems

Problem 1: Text Generation

> **Prompt**: " Act as a graduate-level data science student. Write a 200-word summary of prompt engineering in the first-person active voice. Explain what prompt engineering is, why it is important when working with large language models, and how it improves the quality of AI-generated responses. Use clear academic language. Include APA-style in-text citations and a reference list using credible sources published within the last five years."
>
> **Interface:** CoPilot

Response:

Prompt engineering is an essential skill when working with large language models (LLMs), and I view it as the process of designing effective instructions that guide AI systems toward producing accurate, relevant, and useful outputs. I use prompt engineering to clearly communicate my expectations to a model, including context, desired format, tone, and constraints. By carefully crafting prompts, I can improve the quality and consistency of responses while reducing ambiguity and errors (OpenAI, 2024).

I have learned that prompt engineering is important because LLMs generate responses based on patterns in data and the instructions they receive. When I provide clear and specific prompts, the model is more likely to produce outputs that align with my objectives. Techniques such as role prompting, few-shot prompting, chain-of-thought reasoning, and output formatting help me obtain more reliable results for tasks such as writing, summarization, analysis, and problem-solving (White et al., 2024).

I also recognize that prompt engineering enhances the effectiveness of AI-generated content by improving accuracy, relevance, and structure. Instead of relying on generic responses, I can guide the model to generate information that meets particular academic, professional, or technical requirements. As AI adoption continues to grow across industries, I believe prompt engineering will remain a critical competency for maximizing the value of LLMs while supporting responsible and efficient AI use.

References suggested by model

OpenAI. (2024). *Prompt engineering guide*. OpenAI. https://platform.openai.com/docs/guides/prompt-engineering

White, J., Fu, Q., Hays, S., Sandborn, M., Olea, C., Gilbert, H., Elnashar, A., Spencer-Smith, J., & Schmidt, D. C. (2024). *A prompt pattern catalog to enhance prompt engineering with ChatGPT*. *arXiv*. https://arxiv.org/abs/2302.11382

Problem 2: Sentiment Analysis

Prompt: "Analyze the sentiment of the statement "That was FINE example of teaching an intelligent automation class. :-/". Identify the emotions, explain the implied meaning, and provide a confidence level for each detected emotion and highlight the primary imotion."

Interface: Claude

Response:
Sentiment Analysis

Statement: "That was FINE example of teaching an intelligent automation class. :-/"

Overall sentiment: Negative (mild to moderate), with about 80% confidence.

Key signals

- **Capitalized "FINE":** The emphasis reads as sarcastic stress. Someone sincerely praising the class would more likely write "great" or "excellent," or wouldn't capitalize at all.
- **The ":-/" emoticon:** This signals skepticism, disappointment, or being unimpressed. It contradicts a literal positive reading of "fine."
- **"Fine" itself:** Even without sarcasm, "fine" is faint praise, meaning merely acceptable.
- **The topic:** The phrase "intelligent automation class" may add irony, as if the class fell short of the standard its own subject implies (for example, feeling scripted, mechanical, or unengaging). This is a possible layer, not a certain one.

Emotions detected

Emotion	Confidence	Basis
Disappointment (primary)	~70%	The ":-/" face and the gap between what was expected and what happened
Sarcasm / irony (as a tone)	~65%	The all-caps "FINE" paired with a negative emoticon
Skepticism / being unimpressed	~60%	Lukewarm wording and the ambivalent face
Mild annoyance or frustration	~40%	The emphasis suggests some irritation, but it's understated

Emotion	Confidence	Basis
Sincere mild approval	~20%	"Fine" can mean "okay," but the other cues largely rule this out

Primary emotion: Disappointment

The writer seems to have expected more from the class and is expressing letdown rather than anger. The tone is dry and understated, not hostile.

Implied meaning

The most likely reading is: *"That class was underwhelming, and I'm politely (or wryly) saying so."* The speaker avoids an outright insult and lets the sarcasm and emoticon carry the criticism. It could be aimed at the instructor, the content, or the delivery.

Caveats

Text-only sarcasm is hard to judge with certainty. If the same person habitually types in caps for emphasis, or if "FINE" was a typo or shorthand for "fine" (as in good), the sentiment could be neutral to mildly positive. The missing article ("a fine example") also suggests a quick, informal message, which makes the ":-/" the strongest clue.

Problem 3: Simple Math

Prompt: "Solve the following arithmetic problem step-by-step internally. Add 2,292.32 and 4494 1/3. Convert the final answer into an improper fraction. Return only the fraction with no explanation, words, or additional formatting."

Interface: ChatGPT

Response:

508999/75

Results and Evaluation

Problem 1: Text Generation

The text generation prompt used role prompting, audience specification, formatting requirements, and citation constraints. By instructing the model to act as a graduate-level data science student and write in the first-person active voice, the generated content closely aligned with the expected academic tone. The response accurately explained prompt engineering, its importance in interacting with LLMs, and how it improves response quality through techniques such as role prompting, few-shot examples, and chain-of-thought reasoning. Furthermore, the model successfully incorporated APA-style citations and a reference list, demonstrating its ability to follow complex formatting instructions. This result illustrates how detailed prompts can significantly improve output relevance, structure, and completeness.

Problem 2: Sentiment Analysis

The sentiment analysis prompt focused on emotion detection, sarcasm recognition, and contextual interpretation. Claude correctly identified the overall sentiment as negative and recognized that the capitalization of the word "FINE" and the ":-/" emoticon indicated sarcasm and disappointment rather than genuine praise. The response further classified emotions such as disappointment, skepticism, irony, and mild frustration while providing confidence levels for each finding. The model also acknowledged uncertainty through caveats, demonstrating an understanding of the limitations associated with textual sentiment interpretation. This result highlights the value of using explicit instructions when a task requires nuanced reasoning and contextual analysis beyond simple positive or negative sentiment classification.

Problem 3: Simple Math

The mathematical reasoning prompt combined calculation requirements with strict output constraints. The model successfully followed the instructions and returned only the fractional result, demonstrating effective adherence to the requested output format. The prompt utilized reasoning guidance while controlling the final response format, reducing unnecessary explanations and focusing solely on the desired answer. This demonstrates how prompt engineering can improve precision and consistency in analytical and computational tasks.

Conclusion

This exercise demonstrated the importance of prompt engineering across multiple LLM use cases, including text generation, sentiment analysis, and mathematical reasoning. The results showed that carefully crafted prompts can significantly improve response quality by reducing

ambiguity, increasing accuracy, and providing greater control over output formatting. Role prompting enhanced text generation quality, context-aware instructions improved sentiment analysis, and strict output constraints ensured precise mathematical responses. As organizations increasingly adopt AI technologies, prompt engineering will remain a critical skill for effectively leveraging LLM capabilities while maximizing reliability, consistency, and value from AI-assisted solutions.

References

OpenAI. (2024). *Prompt engineering guide*. OpenAI. https://platform.openai.com/docs/guides/prompt-engineering

White, J., Fu, Q., Hays, S., Sandborn, M., Olea, C., Gilbert, H., Elnashar, A., Spencer-Smith, J., & Schmidt, D. C. (2024). *A prompt pattern catalog to enhance prompt engineering with ChatGPT*. *arXiv*. https://arxiv.org/abs/2302.11382